

Risk Does Not Specify Intervention: Exact-State Diagnostics for an OpenVLA–LIBERO Safety Case Study

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Abstract

Runtime safety for vision-language-action policies is often formulated as predicting whether nominal behavior will catastrophize. That binary label does not identify whether intervention improves realized utility, which finite option is best, or whether a statewise opportunity can be reached sequentially. We study these distinctions in one frozen OpenVLA–LIBERO `glass_recovery` exact-state design. From identical hash-verified branch starts, we execute BASE, a privileged task-preserving DETOUR, and a conservative RETREAT under a common task-success, catastrophe, and safe- noncompletion utility. All 20 sources and 273 eligible decisions are exposed development evidence. Binary Base risk and intervention benefit disagree on 19/273 decisions across 12 sources. Exhaustive tight matching yields three same-risk, different-strict-action pairs across two sources; the exact DETOUR–RETREAT witness has only one independent source. Source-macro utility is 0.2544 for `RISK→BESTFIXED` and 0.5642 for the realized finite-option ORACLE, leaving an observed opportunity of 0.3097. The strongest evaluated OutcomeRouter gains only +0.0178 over `RISK→BESTFIXED` and no method gate passes; the full tiny advantage-decomposed router reaches only 0.2311. Support is mechanically concentrated: all 23 strict RETREAT states and 51/60 strict DETOUR-or-RETREAT states are glass. At fresh sequential first crossing, the direct selector chooses BASE on 24/24 episodes, recovers 0/8 glass episodes, and misses 2/2 known opportunities. The result is therefore a glass-scoped exact-state diagnostic and negative-result case study, not a successful router or a broad VLA-safety benchmark.

1 Introduction

Consider two saved states from the same source, the same glass condition, the same five-action horizon, and the same binary observation that continuing BASE catastrophizes. In state `43347d99...`, realized outcomes for (BASE, DETOUR, RETREAT) are (catastrophe, task success, safe noncompletion), so DETOUR is uniquely best. In state `f83d0199...`, the outcomes are (catastrophe, catastrophe, safe noncompletion), so RETREAT is uniquely best. A perfect binary risk bit is identical for the pair but cannot specify the action. This witness is deliberately narrow: both states come from one source in one mechanical design. Its value is diagnostic, not universal.

This distinction matters whenever a runtime system can choose among preserving the nominal policy, attempting task-preserving recovery, and stopping or retreating safely. Failure detection asks whether nominal continuation is bad. Intervention asks a different question: whether any available alternative is better, and if so which one. A further gap separates a choice that is clear at an authored saved state from one that a sequential policy can realize at the right time. We write these four targets as

$$\text{Base risk} \neq \text{intervention benefit} \neq \text{best finite option} \neq \text{sequential realization}.$$

We use the frozen CrashBench exact-state corpus to measure those distinctions without introducing a new learned method or collecting another outcome. At each eligible decision, three controllers are executed from the same verified branch start. The resulting finite-option outcomes support a source-aware diagnosis of risk-benefit disagreement, strict action ambiguity, policy regret, and oracle-hybrid attribution. The strongest learned selectors remain null under their frozen gates, and a fresh sequential evaluation collapses to BASE. The appropriate scientific object is therefore the diagnostic protocol and its scoped negative result.

Our contributions are exactly four:

1. an exact-state realized-option protocol that restores a complete simulator/controller/continuation state and executes every member of a finite option set, observing outcomes instead of predicting them with a learned world model;
2. a source-aware risk-regret diagnosis that separately measures Base risk, intervention benefit, strict option support, same-risk action ambiguity, and risk-only policy regret;

3. a fair-baseline null and oracle attribution showing that the evaluated learned selectors do not pass the frozen method gate and that benefit gating accounts for most of the reported oracle-hybrid loss; and
4. a statewise-to-sequential boundary preserving the fresh first-crossing null: statewise option opportunity does not imply reliable sequential realization.

The evidence is one OpenVLA checkpoint, one LIBERO pick-and-place task, and one `glass_recovery` mechanical family. The labels `glass`, `offpath`, and `noglass` are a treatment and matched controls, not three hazard families. We make no claim of a deployable router, broad benchmark, cross-mechanism generality, or reliable sequential intervention.

2 Related Work and Novelty Boundary

VLA policies and failure detection. OpenVLA supplies the frozen policy and LIBERO supplies the manipulation environment used in this case study [3, 4]. SAFE learns scalar failure scores from VLA internal features across tasks and policies [1]; Hide-and-Seek localizes trajectory failure signals from coarse labels [8]. Our result is not a new failure detector. It uses all realized outcomes of a fixed option set at identical saved states to ask what a scalar failure target leaves unidentified.

Runtime verification and selective intervention. CheckVLA uses an action-conditioned world model to verify and rewrite action suffixes [7]. CoWAM selects among shared candidate pools using predicted futures and coordination contracts [6]; this is the nearest selective-intervention comparison. Model-based runtime monitoring with interactive imitation learning anticipates failures and can request oversight [5]. CrashBench neither learns a rollout world model nor proposes a deployment layer. Its narrower distinction is realized, full-information branching for three fixed options from serialized simulator states, followed by source-paired diagnosis. CheckVLA and the monitoring literature have substantially broader sequential evidence.

Safety alignment, steering, and counterfactual language. SafeVLA changes the policy through constrained learning [11]; our OpenVLA checkpoint remains frozen. Mechanistic activation steering can causally change VLA behavior [2], so our historical final-readout steering null is not a general statement about steerability. Counterfactual VLA generates reasoning traces that revise driving meta-actions [9], while counterfactual safe reinforcement learning constrains harm relative to a default policy [10]. Here “counterfactual” means only that each listed controller was actually executed from an identical saved simulator state. We do not claim an observational causal estimator, language self-reflection, or novelty from the term itself.

3 Exact-State Realized-Option Protocol

3.1 System, sources, and eligibility

The frozen capture uses checkpoint revision `962318ce...c873` of the OpenVLA LIBERO-Spatial checkpoint (full identity in Appendix A). Decoding is deterministic (no sampling, one beam), and the rollout seed is 0. The retained corpus has 20 unique `source_state_sha256` values and 273 eligible decisions. Historical split labels are 5 train, 7 calibration, and 8 development sources, but they have all been inspected and reused during the successive audits. We therefore treat all 20 as exposed development and make no confirmatory claim.

Each source contributes matched placements and conditions at remaining-action horizons 40, 30, 20, 10, or 5 when mechanically eligible. Eligibility is resolved before method comparison from Base and mechanical contracts. The independent unit is the source state. Conditions, placements, horizons, branches, and frames are repeated observations nested within a source.

3.2 Branch and restoration contract

Figure 1 summarizes the protocol. The collector rebuilds the exact model XML, restores flattened MuJoCo time, position, and velocity state, restores OSC/controller interpolation state, episode counters, robot buffers, observable timing and cached observations, and restores the explicit policy-boundary observation. Before every option, it requires equality of five SHA-256 identities: simulator state, controller state, the full numeric continuation snapshot, observation dictionary, and model XML. A mismatch fails the decision rather than entering the corpus.

The capture also fixes the rollout seed and uses deterministic policy decoding and deterministic structured controllers. It does not retain a separate per-decision hash of global Python/NumPy/framework RNG bytes. We consequently describe exact restoration at the five verified state boundaries above and do not claim a stronger byte-level RNG restoration audit. This distinction is recorded in the reproducibility statement and appendix.

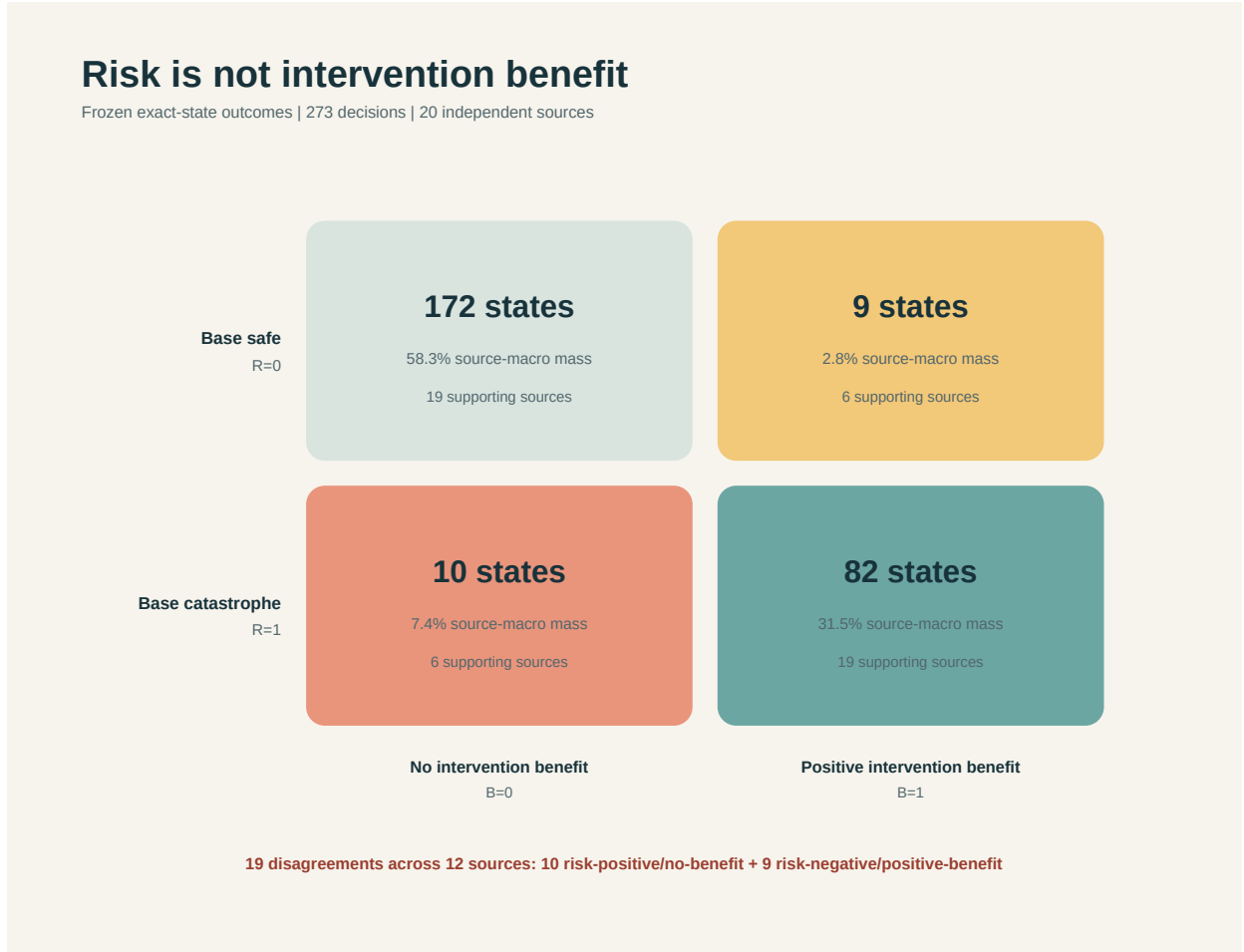
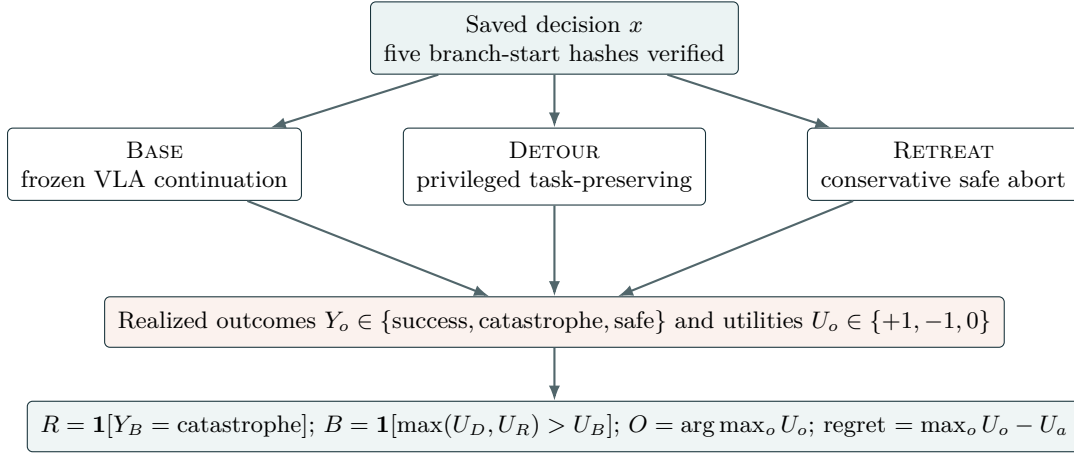


Figure 1: **Exact-state protocol and risk-benefit flow.** Top: every member of the finite option set begins at a hash-verified saved state; the privileged DETOUR uses injected geometry and is not end-to-end learned. Bottom: the complete raw $R \times B$ table over 273 repeated decisions from 20 independent sources. The 10 $R=1, B=0$ decisions and nine $R=0, B=1$ decisions each occur in six sources; their source sets are disjoint. All observations belong to the single `glass_recovery` mechanical family.

3.3 Options, outcomes, and decision targets

BASE continues the frozen OpenVLA policy from the restored observation. DETOUR is a privileged structured controller with access to glass geometry; it is designed to preserve task progress and complete the original placement. RETREAT moves back and up, then holds; it prioritizes avoiding catastrophe but normally does not complete the task. These are fixed option contracts, not a common action budget or a learned proposal set.

Each branch receives exactly one terminal outcome: task success, catastrophe, or safe noncompletion, mapped to utility +1, −1, or 0. For a saved state x , let U_B, U_D, U_R denote the three realized utilities. With the frozen strict benefit epsilon of zero,

$$R(x) = \mathbf{1}[Y_B = \text{catastrophe}], \quad (1)$$

$$B(x) = \mathbf{1}[\max(U_D, U_R) > U_B], \quad (2)$$

$$O(x) = \arg \max_{o \in \{B, D, R\}} U_o, \quad (3)$$

$$A_{\text{int}}(x) = \max(U_D, U_R) - U_B, \quad A_{DR}(x) = U_D - U_R. \quad (4)$$

Oracle ties use the order BASE, DETOUR, RETREAT. Strict support is a different label: it requires a unique maximizer under the frozen support margin $\gamma = 0.5$, excluding all ties.

3.4 Source-aware estimands

Raw decision counts describe repeated observations. Equal-source metrics first compute a rate within each source and then average across sources. Conditional source-macro probabilities average only source conditionals whose denominator is nonzero; they are not ratios of macro joint cells. Learned methods use the frozen leave-one-source-out predictions and their existing foldwise calibration rules. Oracle and oracle-hybrid combinations are diagnostic upper bounds, not deployable methods.

4 Model-Free Risk, Benefit, and Option Evidence

4.1 Risk and benefit disagree in both directions

The raw $R \times B$ table contains 172 (0, 0), 9 (0, 1), 10 (1, 0), and 82 (1, 1) decisions. Thus 19/273 decisions disagree across 12 independent sources. Ten risky Base decisions have no better intervention ($R=1, B=0$), across six sources. Nine non-catastrophic Base decisions are nevertheless improvable ($R=0, B=1$), also across six sources. The two source sets are disjoint. Mechanically, the first direction is concentrated in glass (nine glass and one offpath decision), while the second occurs only in controls (five noglass and four offpath decisions). This supports separation within the design, not a cross-family law.

Raw conditionals are $P(B=1 \mid R=1) = 82/92 = 0.8913$, $P(B=1 \mid R=0) = 9/181 = 0.0497$, and $P(R=1 \mid B=1) = 82/91 = 0.9011$. Their equal-source counterparts are 0.8700 over 20 denominator-bearing sources, 0.0415 over 19, and 0.9328 over 19. The high association explains why risk is a strong baseline; the 19 disagreements establish only that it is not decision-complete.

4.2 Same binary risk, different strict action

We exhaustively form strata matched on source, condition, remaining-action horizon, and binary R , retaining every stratum with at least two different strict optimal labels. This deterministic rule yields exactly three tight pairs across two sources (Table 1); it does not select a visually preferred subset. The only exact DETOUR–RETREAT pair has one-source support.

Condition	Source	H/R	State A: (B, D, R)	State B: (B, D, R)	Strict actions
glass	69fa7e93...	5/1	43347d99...: (cat, success, safe)	f83d0199...: (cat, cat, safe)	Detour / Retreat
offpath	69fa7e93...	5/0	db05c179...: (success, safe, safe)	09da0904...: (safe, success, safe)	Base / Detour
noglass	b830bede...	5/0	7cc66a27...: (success, safe, safe)	2ccd971e...: (safe, success, safe)	Base / Detour

Table 1: **Complete deterministic tight-witness set.** All three pairs are shown. They span 2 independent sources in the one `glass_recovery` mechanical family. The exact Detour/Retreat contrast is supported by 1 source. “offpath” and “noglass” are controls, not families. Full state and observation hashes plus deployable feature summaries are in Appendix D.

5 Fair Baselines and Oracle Attribution

5.1 The learned-method result is null

All comparisons in this section use the same 20 source-held-out prediction blocks and source-macro utility. `RISK→BESTFIXED` chooses both its non-Base option and threshold within each outer calibration fold; it is not a globally fixed Risk-to-Detour rule. It reaches utility 0.2544. The strongest evaluated learned method, OutcomeRouter, reaches 0.2722, a gain of only +0.0178, below the frozen +0.03 requirement. It also misses all three strict Base/Detour/Retreat recall requirements, so no Phase 2.5B method gate passes. The complete tiny nonlinear advantage-decomposed router reaches 0.2311, below the risk reference, and no advantage router passes its method gate.

The null does not erase diagnostic structure. Conditional on a true benefit gate, the tiny nonlinear choice head passes its frozen strict Detour/Retreat point-estimate gate, while both linear heads fail. That head is a capacity diagnostic only: it is not the paper method, not an architecture-selection result, and not evidence that a larger router would solve the benefit gate.

Policy or diagnostic	Utility	Gain vs. risk ref.	Gap to Oracle	Gate/status
<code>RISK→BESTFIXED</code>	0.2544	0.0000	0.3097	strong scalar-risk reference
OutcomeRouter	0.2722	+0.0178	0.2919	no method pass
LearnedGate + OracleChoice (tiny)	0.2714	+0.0169	0.2928	oracle diagnostic
OracleGate + learned linear choice	0.4975	+0.2431	0.0667	oracle diagnostic
OracleGate + learned tiny choice	0.4917	+0.2372	0.0725	capacity diagnostic
Full tiny ADR	0.2311	-0.0233	0.3331	no method pass
ORACLE	0.5642	+0.3097	0.0000	realized upper bound

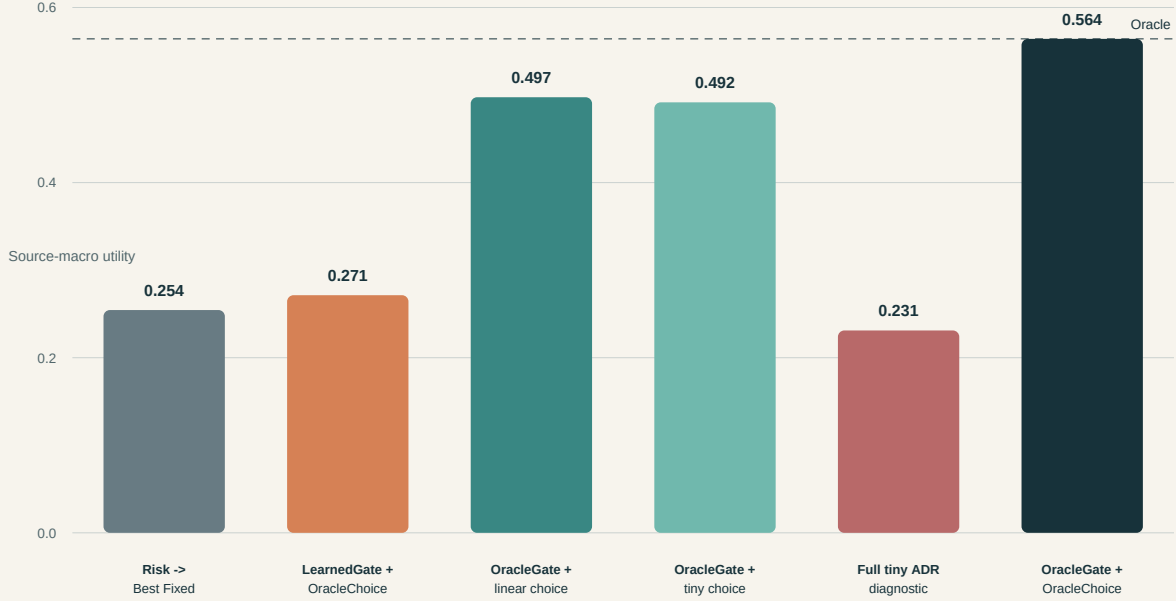
Table 2: **Risk reference, learned selectors, hybrids, and Oracle.** Values are equal-source utility over 20 exposed-development sources and 273 repeated decisions. OracleGate and OracleChoice require realized labels and are diagnostic only. No displayed learned method passes its frozen method gate.

5.2 Benefit gating dominates the reported composition gap

The finite-option ORACLE reaches 0.5642, giving an observed opportunity of $0.5642 - 0.2544 = 0.3097$ above `RISK→BESTFIXED`. For the tiny candidate, OracleGate plus learned choice loses 0.0725 from Oracle and recovers 0.7659 of that opportunity. LearnedGate plus OracleChoice loses 0.2928 and recovers only 0.0547. The full composition recovers -0.0753 . In this frozen composition, benefit gating is the dominant empirical loss. This is attribution using unavailable oracle components, not a proposal for deployment.

The learned benefit gate dominates the frozen oracle gap

Tiny-choice loss = 0.0725 | learned-gate loss = 0.2928



Oracle combinations are diagnostic only. The tiny nonlinear head is a capacity diagnostic, not the paper method.

Figure 2: **Oracle-gap attribution.** Source-macro utilities use the same 20 exposed-development sources. The 0.3097 opportunity is the observed difference between RISK→BESTFIXED (0.2544) and the realized finite-option Oracle (0.5642). Every OracleGate or OracleChoice bar is diagnostic only. The tiny nonlinear head is a conditional capacity diagnostic, not a successful router.

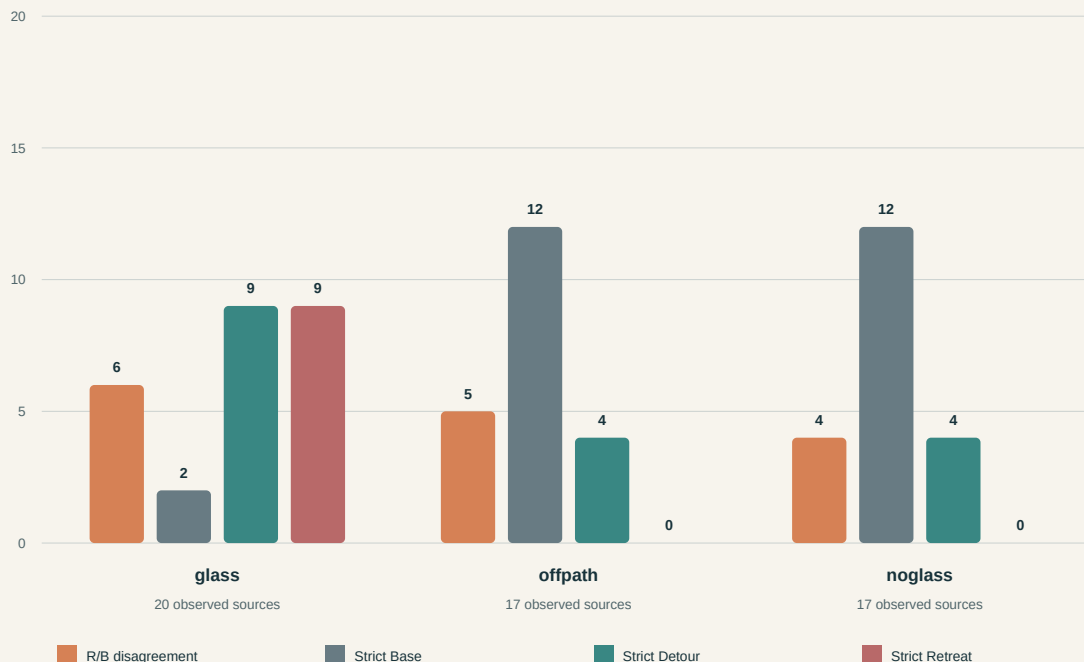
6 Source and Mechanical-Scope Audit

Nineteen of 20 sources contain both benefit labels. Strict support is present for Base, Detour, and Retreat in 16, 13, and 9 sources, respectively. However, pooled source counts conceal mechanical concentration. Glass contains 101 decisions from 20 sources, offpath 86 from 17, and noglass 86 from 17. All 23 strict Retreat states, across nine sources, are glass. Glass also contains 51/60 strict Detour-or-Retreat states across 16 sources; all 60 occur across 17 sources. Non-glass controls contribute nine strict Detour states across six sources and no strict Retreat state.

Five sources contain both a strict Detour and a strict Retreat label when conditions are pooled, but only two contain such a flip within glass. The tight exact Detour/Retreat witness in Table 1 has one source. These nested counts are not interchangeable. They support a glass-scoped diagnosis while ruling out a claim of broad option ambiguity.

Source support is real, but mechanically glass-scoped

All three condition labels share one glass-recovery design; all 23 strict Retreat states and 51/60 strict D/R states are glass



Offpath and noglass are matched controls, not independent mechanical hazard families. Glass strict D/R: 51 states / 16 sources; all strict D/R: 60 / 17.

Figure 3: **Condition-level source support within one mechanical family.** The 20-source universe has condition-specific effective counts of 20/17/17 sources for glass/offpath/noglass, respectively. The bars show effective independent-source counts separately for the **glass**, **offpath**, and **noglass** conditions. All conditions share the single **glass_recovery** family; offpath and noglass are matched controls. Glass supplies all 23 strict Retreat states (nine sources) and 51/60 strict Detour-or-Retreat states (16 of the 17 sources supporting any strict D/R state).

7 From Statewise Opportunity to Sequential Realization

Statewise exact branching gives labels at authored states. A sequential policy must reach those states from reset and cross a decision boundary at a useful time without unnecessary control interventions. The frozen sequence in Table 3 is therefore a boundary analysis, not a second method arc.

Matched-state source-OOF evaluation contains a large Oracle opportunity, yet OutcomeRouter gains only +0.0178 and no method passes. On four development sources, P2 changes 7/12 successes and 4/12 catastrophes to 8/12 and 3/12 at 2/12 interventions, while missing both known T-20 recoveries. Simple temporal summaries in P2.5 do not repair the ordering. P3.1 then obtains extremely high offline source-held-out separation (AUC 1.000 for recovery-open versus hard control and 0.995 for intervention-needed versus hard control), but only 0.632 source-macro accuracy.

The decisive fresh P3.2 closeout uses eight sources and three conditions per source. The direct selector chooses BASE in all 24 episodes. It therefore has zero false interventions by inactivity, not by successful selectivity. It recovers 0/8 glass episodes and misses 2/2 known recovery opportunities. The older P2 comparator selects Detour in 5/24, recovers 1/8 glass episodes, and retains 16/16 control successes. Neither result authorizes reliable sequential intervention, and the frozen threshold is not reopened.

Stage	Evidence	Sources/units	Scoped interpretation
Matched state	Oracle 0.5642; risk ref. 0.2544; OutcomeRouter 0.2722	20/273	opportunity exists; learned gain +0.0178; no pass
P2	7/12 to 8/12 success; 4/12 to 3/12 catastrophe; 2/12 intervention	4/12	development only; misses 2/2 known recoveries
P2.5	raw max AUC 0.357; MA-3/5/8 AUC 0.286	4/11	tested simple temporal summaries do not repair ordering
P3.1	recovery-open AUC 1.000; intervention-needed AUC 0.995; accuracy 0.632	9/142	offline source-held-out separability only
P3.2	Direct: Base 24/24; glass recovery 0/8; known misses 2/2	8/24	fresh first-crossing collapse to Base

Table 3: **Statewise-to-sequential evidence.** The fresh closeout has 8 sources. Sources are independent; units are repeated decisions, anchors, or episodes as indicated. The P3.2 negative closeout is one fresh eight-source cohort in the same `glass_recovery` design, not evidence that all sequential methods are impossible.

8 Limitations and Safety Scope

Scope and statistical unit. This is one frozen OpenVLA checkpoint, one LIBERO task, one embodiment, and one glass-recovery mechanical design. All 20 sources are exposed development. Decision counts are repeated observations; source counts are the inferential unit. The three condition labels are not independent hazard families. Exact Detour/Retreat ambiguity has only one-source support.

Options and utility. The result is conditional on exactly three engineered options and the frozen $+1/0/-1$ utility. Privileged Detour uses geometry unavailable to a generic deployment. Retreat is a conservative safe-abort controller rather than a task recovery policy. The utility does not charge different duration, path length, latency, or control effort, so Oracle is only the best realized option under this contract.

Restoration and artifact availability. The collector verifies five branch-start hashes, but does not record a separate per-decision global RNG-state hash. Deterministic decoding and fixed seed limit the practical dependence without proving byte-identical RNG restoration. Raw capture files are locally hash-pinned but ignored by Git; the reviewed PIVOT tables, figures, and manifests are tracked. Raw witness pixels were not retained, so the released witness table provides observation hashes and feature summaries rather than images.

Learning and sequential deployment. No learned method passes the frozen method gate. The tiny nonlinear choice result is conditional on an oracle gate. High offline P3.1 AUC does not transfer to fresh first crossing. The P3.2 null closes this protocol, not every possible temporal model. We do not propose threshold lowering, another router, or a new cohort from these results.

Ethics and safety. All claim-bearing outcomes are simulated robot manipulation; no human-subject or personal data are used. The diagnostic could help expose unsafe assumptions in runtime intervention systems, but it is not a safety certificate. Reporting the null method and sequential results reduces the risk that an offline oracle gap is mistaken for deployable protection. Any real-robot use would require independent mechanical validation, calibrated monitoring, fail-safe hardware, and task-specific oversight beyond this study.

9 Conclusion

Within one frozen OpenVLA–LIBERO glass-recovery design, realized exact-state outcomes show that Base catastrophe status does not uniquely determine intervention benefit or the best finite option. The phenomenon is supported by 12 sources for risk-benefit disagreement but by only two sources for tight same-risk action pairs, with one source for the exact Detour/Retreat witness. A substantial observed Oracle opportunity remains, yet the evaluated learned selectors do not pass their method gates and the fresh sequential selector collapses to Base. The defensible result is a source-aware, glass-scoped diagnostic and negative-result case study. Broader claims require

separately authorized evidence across mechanical families, tasks, policies, and confirmatory sources; they are not inferred here.

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A Frozen Corpus and Provenance

Table 4 records the contracts used by every main-text number. The PIVOT-0 synthesis generated no rollout and trained no model; it reads frozen Phase 2.5A, Phase 2.5B, Phase 2.5B-R, and sequential-closeout artifacts. Historical train/calibration/development names remain provenance fields only. PIVOT-0 treats all 20 sources as exposed development.

Contract	Frozen value
Policy checkpoint	<code>openvla/openvla-7b-finetuned-libero-spatial</code> , revision <code>962318cec55ac10993ff0f5f43eda9a270b4c873</code>
Capture commit	<code>d4751330395eec11691f4830761b76f5f1ac36b7</code> (clean)
Capture protocol hash	<code>b487d8b5cc20c1fe932d8ee4ff67f05883e2b010b6ee9eb155eb9b93c2d1a4c3</code>
PIVOT generation commit	<code>98b3ffe0aec2838448d0536b6c62036e37487f6f</code> (clean)
PIVOT evidence fingerprint	<code>dc48ff317dad</code> ; capture fingerprint <code>d4751330395e</code>
Sources and decisions	20 exposed-development sources; 273 eligible decisions; 819 realized option rows
Historical split provenance	5 train, 7 calibration, 8 development sources; no confirmatory split in this synthesis
Mechanical scope	one <code>glass_recovery</code> family; <code>glass</code> treatment plus <code>offpath/noglass</code> controls
Options	<code>base_continue</code> , privileged <code>detour_complete</code> , <code>retreat_hold</code>
Outcomes and utility	task success = +1; safe noncompletion = 0; catastrophe = −1
Decode and seed	deterministic decoding (no sampling, one beam); rollout seed 0
PIVOT actions	new rollouts false; new training false; fresh test outcomes loaded false

Table 4: Frozen corpus and provenance contract. Effective independent sources: 20; repeated decisions: 273.

B Exact Restoration, Eligibility, and Finite-Option Semantics

Saved state. The collector stores flattened MuJoCo time, *qpos*, and *qvel*; exact model XML; OSC goals, gains, torques, and interpolator state; integration inputs; episode counters; robot temporal buffers; observable values and timers; the observation cache; and the explicit policy-boundary observation. The historical name “controller state” therefore denotes a broader continuation snapshot, not only an OSC command.

Five required identities. Before each option branch, the collector rebuilds the model, restores the numeric state and continuation snapshot, and recomputes SHA-256 identities for (i) simulator state, (ii) controller/dynamics state excluding scene-specific observations, (iii) the complete numeric continuation including observation history, (iv) every policy-boundary observation field with dtype and shape, and (v) model XML. Any mismatch raises `exact_state_restore_mismatch` and the decision cannot be retained. The 273 retained decisions each have three completed option rows under this fail-closed code path.

RNG boundary. The run fixes seed 0 and deterministic policy generation, and the structured controllers are deterministic. The stored branch identities do not include a separate serialization or hash of global Python, NumPy, or framework RNG state at every decision. The exact-state claim in this paper is restricted to the five verified branch-start identities. A broader benchmark should add explicit RNG serialization even when decoding is deterministic.

Eligibility. Source/placement eligibility is mechanical and Base-defined, not selected by a router result. Exclusions include invalid initial catastrophe, on-path failure to produce the intended catastrophe, catastrophe before the requested horizon, and a control terminating before the matched anchor. A placement/condition/ horizon enters only when a valid branch start exists. All retained branches receive exactly one of the three terminal outcomes. The PIVOT synthesis keeps all 273 retained decisions and does not recensor after observing option labels.

Interpretation. Executing all three branches is full information for this finite option set at these saved states. It does not identify the value of an unexecuted controller, an observational population effect, or the best policy in a continuous action space. Privileged Detour receives obstacle geometry. Retreat is a conservative fallback. Their different control horizons and effort are not charged by the frozen utility.

C Complete Risk-Benefit Accounting

Cell	Raw decisions	Supporting sources	Equal-source joint mass
$R = 0, B = 0$	172	19	0.5831
$R = 0, B = 1$	9	6	0.0278
$R = 1, B = 0$	10	6	0.0742
$R = 1, B = 1$	82	19	0.3150
Total	273	20	1.0000

Table 5: Raw and equal-source $R \times B$ accounting over 20 independent sources. A supporting-source count means that the source contains at least one decision in the cell; counts therefore do not sum to 20.

The raw conditional probabilities are 0.8913 for $P(B = 1 \mid R = 1)$, 0.0497 for $P(B = 1 \mid R = 0)$, and 0.9011 for $P(R = 1 \mid B = 1)$. Equal-source conditional means are respectively 0.8700 over 20 denominator-bearing sources, 0.0415 over 19, and 0.9328 over 19. Ratios of the equal-source joint cells in Table 5 are different estimands and are not substituted for these conditional means.

The 10 $R = 1, B = 0$ decisions occur at horizons 5/10/30/40 in counts 6/1/1/2 and conditions glass/offpath in counts 9/1. The nine $R = 0, B = 1$ decisions occur at horizons 5/10/20/30 in counts 5/2/1/1 and conditions noglass/offpath in counts 5/4. Each direction has six sources, and the source sets are disjoint.

D Complete Tight Witnesses

The deterministic witness rule matches source, condition, horizon, and binary risk exactly. Table 6 contains every resulting pair. The “feature summary” columns are computed from deployable robot state, nominal action, and frozen hidden features. Raw pixels are not retained; exact observation and hidden hashes make that limitation explicit.

Table 6: Full deterministic witness records: 3 pairs across 2 independent sources, with 1 source for the exact Detour/Retreat contrast.

Field	Value
Pair 1 stratum	mechanical family glass_recovery; condition glass; source 69fa7e93588c6aea4b134efa44633d38556ce1656b46df276021920a664881a3; $H = 5$; $R = 1$
Pair 1 states	A 43347d99e35479cdc3d3197e5c6127d8bb063ec834656eecd505a39f52d7167; B f83d0199d1f4fdc40ae28dc3a5c02514dc5ae737d391602dea012bb750f98642
Pair 1 observations	A fe26c03740a7719695c9e5e2bdbac19a64e0bcb93c52ce587fb187c95f55652e; B cd048daaae15635b63b6ea1c4568163ee75066ca3ae1eccb7cb69c2323750e64
Pair 1 out-comes/utility	A: (Base catastrophe -1 , Detour success $+1$, Retreat safe 0), strict Detour. B: (Base catastrophe -1 , Detour catastrophe -1 , Retreat safe 0), strict Retreat.
Pair 1 feature summary	z-scored robot/action distance 3.2781; hidden cosine distance 0.6481; hidden hashes A 9810a1fd95eb8fb0b82787c1f52adf76c98e2f27c1683fa32b238eb2c60955ad, B 334985f6df5e8d698a1d0e06d3a8dabb9c04d556257be6a8f5d33c7b2b94de3.
Pair 2 stratum	mechanical family glass_recovery; condition offpath; source 69fa7e93588c6aea4b134efa44633d38556ce1656b46df276021920a664881a3; $H = 5$; $R = 0$
Pair 2 states	A db05c17974a1032f5835681f51f63fcb978004eecfe48098e0892f5bd6ada082; B 09da0904e292712e5ed0fbdeae877e637acabfb5f929d8276f5fdb0c3004a96
Pair 2 observations	A 226d3ac32a8c77573e890d4948eb07e0fffd911eddb26ee53763ae5ab885734b7; B 910cb06477e5969823bbb792a43446d4c99146799c761208495a74730aabf11a
Pair 2 out-comes/utility	A: (Base success $+1$, Detour safe 0 , Retreat safe 0), strict Base. B: (Base safe 0 , Detour success $+1$, Retreat safe 0), strict Detour.
Pair 2 feature summary	z-scored robot/action distance 3.8686; hidden cosine distance 0.4464; hidden hashes A 4c1b430253f5e6f64a19ea060abb815b8ffa91a1e675b1649466d6b0c2832f35, B 5a148ac7f5e7145c671ab2eb0996613473be787531dc17889ecacc5d1a8a617e.
Pair 3 stratum	mechanical family glass_recovery; condition noglass; source b830bede73f505b84b88a6596e92c22a5a4c0ded1e275e8e5b28a599849d2df6; $H = 5$; $R = 0$

Field	Value
Pair 3 states	A 7cc66a27ce2971b4dd170aa8f8b415751cf10acaf0a4eb4a2ac10682c2b9dda6; B 2ccd971e77ce745c2eb16199826277fcd3c3072243e5e2d9b5214fb4b1d0b89c
Pair 3 observations	A 44fadc36ca73e1c03ec22f5653aed6748c82f7bc39c7ce1edfc8fc4c7813faa1; B 7dd048488f8e3683036bd9e47426b4b98b05fffb109f09eaa99d814ccca3f07fa
Pair 3 out-comes/utility	A: (Base success +1, Detour safe 0, Retreat safe 0), strict Base. B: (Base safe 0, Detour success +1, Retreat safe 0), strict Detour.
Pair 3 feature summary	z-scored robot/action distance 2.0406; hidden cosine distance 0.6742; hidden hashes A c9370f4f86e582a51e3fc1acc5023bf8854e57817c7196dca65773a31a98ae85, B 97d05dc72f018a054679495cfd8d7584f80389153a74862f086099ce9ec99c26.

Pair 1 is the only exact Detour/Retreat contrast and has one independent source. Pairs 1 and 2 share that source, so the complete three-pair set has two independent sources, not three.

E Complete Source-Macro Policy-Regret Table

Policy	Success	Cat.	Safe	Interv.	Utility	Regret	Opp. rec.
Always Base	.5197	.3892	.0911	.0000	.1306	.4336	-.4000
Always Detour	.4347	.1800	.3853	1.0000	.2547	.3094	.0009
Always Retreat	.0000	.1858	.8142	1.0000	-.1858	.7500	-1.4215
Risk → Detour	.4989	.2511	.2500	.4981	.2478	.3164	-.0215
Risk → Retreat	.4394	.3097	.2508	.3506	.1297	.4344	-.4027
Risk → Best Fixed	.5022	.2478	.2500	.4947	.2544	.3097	.0000
Oracle Benefit Gate → fixed Detour	.6383	.1483	.2133	.3428	.4900	.0742	.7605
OracleGate + tiny learned choice	.6158	.1242	.2600	.3428	.4917	.0725	.7659
Tiny LearnedGate + OracleChoice	.4878	.2164	.2958	.4567	.2714	.2928	.0547
Full tiny ADR	.4733	.2422	.2844	.4567	.2311	.3331	-.0753
Oracle	.6383	.0742	.2875	.3428	.5642	.0000	1.0000

Table 7: All policy-regret rows under the same 20 sources and 273 repeated decisions under the source-macro contract. “Opp. rec.” is the fraction of the Oracle improvement above Risk → Best Fixed. Negative values mean a policy falls below that risk reference. Oracle gates/choices are diagnostic and unavailable at deployment.

The Phase 2.5B OutcomeRouter row is not repeated in Table 7 because the frozen PIVOT policy-regret CSV records the fixed/risk/oracle-hybrid composition table. Under the same 20 source blocks, OutcomeRouter has success 0.5008, catastrophe 0.2286, intervention 0.4831, and utility 0.2722. Its +0.0178 utility gain over the 0.2544 risk reference is below the +0.03 gate and it fails the three strict-recall criteria.

F Learned Selector and Attribution Audits

Method	Utility	Base rec.	Detour rec.	Retreat rec.	Frozen result
Risk → Best Fixed	.2544	.5595	.5705	.0556	reference
OutcomeRouter	.2722	.5926	.3269	.3704	no pass
Support-balanced OutcomeRouter	.2672	.6534	.3013	.4815	no pass
Hidden-only OutcomeRouter	.2644	.4881	.4359	.5185	no pass
ADR linear full	.2097	.5874	.3910	.2593	no pass
ADR linear split	.2222	.4999	.4487	.3704	no pass
ADR tiny nonlinear choice	.2311	.5749	.4551	.4444	no pass

Table 8: Frozen source-OOF learned-selector results over 20 sources. Phase 2.5B is **INCONCLUSIVE** and fail-closed; Phase 2.5B-R records **CHOICE_CAPACITY_BOTTLENECK** while every complete ADR has **method_pass=false**.

On the 60 strict Detour/Retreat decisions, conditional balanced accuracy is 0.5579 for the full linear head, 0.6369 for the split linear head, and 0.7253 for the fixed tiny nonlinear head. Their Detour/Retreat recalls are 0.7421/0.3738, 0.8354/0.4385, and 0.8559/0.5947. Only the tiny conditional head passes the point-estimate choice gate. All candidates share a weak learned benefit score (AUROC 0.6062, AUPRC 0.4783), and no full composition passes.

For the tiny candidate, the source-macro losses from Oracle are 0.0725 for OracleGate plus learned choice, 0.2928 for LearnedGate plus OracleChoice, and 0.3331 for the full composition. The split-linear OracleGate hybrid reaches 0.4975, slightly above the tiny hybrid’s 0.4917. The tiny head is therefore not selected as a generally superior model; it diagnoses conditional capacity only.

G Condition and Horizon Support

Condition	Decisions	Sources	$R \neq B$	$R \neq B$ src.	Strict B	Strict D	Strict R	D/R ties
glass	101	20	9	6	2	28	23	31
offpath	86	17	5	5	24	4	0	0
noglass	86	17	5	4	35	5	0	0

Table 9: Condition accounting over a 20-source universe inside the single **glass_recovery** mechanical family; glass/offpath/noglass contribute 20/17/17 sources. Here B/D/R in the strict columns denote Base, Detour, and Retreat, not the binary benefit variable.

Across all conditions, strict Base/Detour/Retreat counts are 61/37/23 and their source supports are 16/13/9. Horizon-level strict Base/Detour/Retreat counts are 4/2/1 at H40, 15/6/1 at H30, 16/8/5 at H20, 11/11/9 at H10, and 15/10/7 at H5. Retreat support is both glass-only and late-state concentrated: 16/23 strict Retreat decisions occur at H5 or H10.

Nineteen sources contain both benefit labels. Nine contain a strict Base/Detour flip, nine a strict Base/Retreat flip, and five a pooled strict Detour/Retreat flip. Only two have a within-glass strict Detour/Retreat flip. The complete per-source flag matrix is the tracked **source_support.csv**; it is not expanded into 20 rows here because the artifact map identifies the exact file and SHA-256.

H Expanded Sequential Boundary

Stage	Method/diagnostic	Sources	Units	Success	Interv.	Frozen reading
P2	exact T-20 opportunity	4	4	2 known	–	option label, not policy
P2	fixed T-20 Router	4	12	7/12	0/12	misses 2/2 known opportunities
P2	dynamic first crossing	4	12	8/12	2/12	3/12 catastrophes; development only
P2.5	trajectory morphology	4	11	–	–	raw max AUC .357; moving-average AUC .286
P3.1	direct head strict LOSO	9	142	–	–	recovery AUC 1.000; needed AUC .995; source-macro accuracy .632
P3.2	old P2 comparator	8	24	17/24	5/24	recovers 1/8 glass; two unnecessary control interventions
P3.2	direct recovery Router	8	24	16/24	0/24	Base 24/24; recovers 0/8 glass; misses 2/2
P3.2	exact T-20 Detour	8	8	2/8	–	option diagnostic; not deployed timing
P3.2	trajectory maxima	2 positive	24	–	0/24	known-recovery vs control AUC .250

Table 10: Complete frozen statewise-to-sequential summary over stage-specific sets of 4, 9, and 8 sources. “Units” are repeated decisions, anchors, or episodes, not additional independent sources.

The direct P3.2 method’s zero false interventions are mechanically coupled to choosing Base in every episode. They are not evidence of a selective policy that both preserves controls and intervenes on opportunities. The old P2 comparator retains task success on all 16 controls but recovers only one of eight glass episodes. These results close threshold and temporal rescue for the frozen protocol; they do not prove that sequential intervention is impossible.

I Artifact Availability and Reproduction Boundary

The reviewed PIVOT manifest, seven CSV tables, story decision, and three PDF figures are tracked in Git and hash-pinned by the PIVOT manifest. The capture manifest, feature archive, decision metadata, and option-rollout JSONL are present in the working environment and match the hashes recorded by PIVOT, but the **results/counterfactual_router/** directory is ignored by Git. A clean repository checkout therefore supports auditing all paper numbers from tracked reviewed summaries, but does not by itself contain the raw hidden features or 819 branch rows. We call this **local_untracked_hash_pinned**, not repository-tracked availability.

The root ICLR manifest predates PIVOT-0 and does not enumerate the later PIVOT package. We preserve that frozen manifest and add the publication artifact map instead of rewriting history. The PIVOT manifest pins its outputs but cannot self-pin its own bytes; the publication artifact map records the current manifest hash separately. Generation commits and later record commits are kept as distinct fields.

The local package audit verifies PIVOT required-artifact membership and hashes, all headline values, source/family language, prohibited affirmative claims, artifact-map schema, PDF existence, and the absence of unfinished markers. Repository audit and unit tests remain separate checks. None of these commands opens a simulator, fits a model, or contacts Quest.

J Non-Glass v1 Exclusion

The optional unstable-placement v1 job is not scientific evidence for this paper. It attempted eight nominal source candidates, marked two as selected before location resolution, resolved zero physical blocks, opened zero option decisions, performed zero restoration audits, and produced zero ordinary option-outcome rows. Its terminal status is `INVALID_PREFLIGHT_ZERO_OPTION_OUTCOMES`. The failure was an engineering geometry/source-freeze defect, not a positive or negative result about non-glass intervention ambiguity. v1 contributes nothing to the 20-source or 273-decision denominators and is mentioned here only to make exclusion auditable.

K Claim Boundary Checklist

Supported within scope	Not established
Exact realized outcomes distinguish Base risk, intervention benefit, best finite option, and sequential realization in one frozen design.	A universal theorem about risk, regret, hazards, tasks, backbones, or controllers.
Risk and benefit disagree on 19/273 decisions across 12 sources.	A broad, confirmatory VLA-safety benchmark.
Three exact tight pairs occur across two sources; the D/R witness has one source.	Reliable non-glass Retreat support or replicated cross-mechanism option ambiguity.
Risk $\rightarrow$ Best Fixed leaves an observed 0.3097 finite-option Oracle opportunity.	A reachable or learned Oracle at deployment.
Benefit gating dominates the reported oracle-hybrid loss.	A successful, superior, or deployable learned router.
Fresh direct first crossing collapses to Base in this protocol.	The impossibility of all temporal or sequential intervention methods.

Table 11: Submission-facing claim boundary. The core corpus has 20 sources; the tight witness set has 2 sources and its exact Detour/Retreat contrast has 1.